

HTTP Status Codes

Express & Node.js Reference

1xx — Informational

100 Continue

Server got the headers; client can send the body.
`// handled by HTTP layer automatically`

101 Switching Protocols

Server agrees to switch protocol (e.g. WebSocket upgrade).
`// used internally by ws / socket.io`

2xx — Success

200 OK

Standard success. Use for GET, PUT, PATCH.
`res.status(200).json({ data });`

201 Created

New resource created. Use after POST.
`res.status(201).json({ id, message: 'Created' });`

202 Accepted

Request received but processing not done yet.
`res.status(202).json({ jobId });`

204 No Content

Success, nothing to return. Use for DELETE.
`res.status(204).send();`

3xx — Redirection

301 Moved Permanently

Resource permanently moved. Browser caches this.
`res.redirect(301, 'https://new-url.com');`

302 Found

Temporary redirect. Browser does not cache.
`res.redirect('/login'); // default`

304 Not Modified

Cached version is still valid, skip the body.
`// handled via ETag / Last-Modified headers`

307 Temporary Redirect

Temp redirect, keeps original HTTP method.

```
res.redirect(307, '/new-path');
```

4xx — Client Errors

400 Bad Request

Invalid syntax or missing required fields.

```
res.status(400).json({ error: 'Bad input' });
```

401 Unauthorized

Not logged in. Token missing or invalid.

```
res.status(401).json({ error: 'Login required' });
```

403 Forbidden

Logged in but not allowed. Wrong role/permission.

```
res.status(403).json({ error: 'Forbidden' });
```

404 Not Found

Resource does not exist.

```
res.status(404).json({ error: 'Not found' });
```

409 Conflict

Duplicate data conflict (e.g. email already used).

```
res.status(409).json({ error: 'Already exists' });
```

422 Unprocessable Entity

Syntax OK but validation failed.

```
res.status(422).json({ errors: result.array() });
```

429 Too Many Requests

Rate limit hit. Client is sending too fast.

```
res.status(429).json({ error: 'Slow down' });
```

5xx — Server Errors

500 Internal Server Error

Unexpected server crash. Catch-all error.

```
res.status(500).json({ error: 'Server error' });
```

502 Bad Gateway

Upstream server returned a bad response.

```
// common behind a proxy / load balancer
```

503 Service Unavailable

Server down or in maintenance mode.

```
res.status(503).json({ error: 'Try later' });
```

504 Gateway Timeout

Upstream took too long to respond.

```
// common with slow DB queries behind proxy
```

Quick Tips

- 401 = not logged in | 403 = logged in but not allowed
- 400 = bad syntax | 422 = bad data / validation failed
- 204 has NO body — do not call `res.json()` with it
- Never return 200 with `{ success: false }` — use the right error code
- Global error handler: `app.use((err, req, res, next) => res.status(err.status || 500).json({ error: err.message }))`